Topic: Phase portraits for equal real Eigenvalues

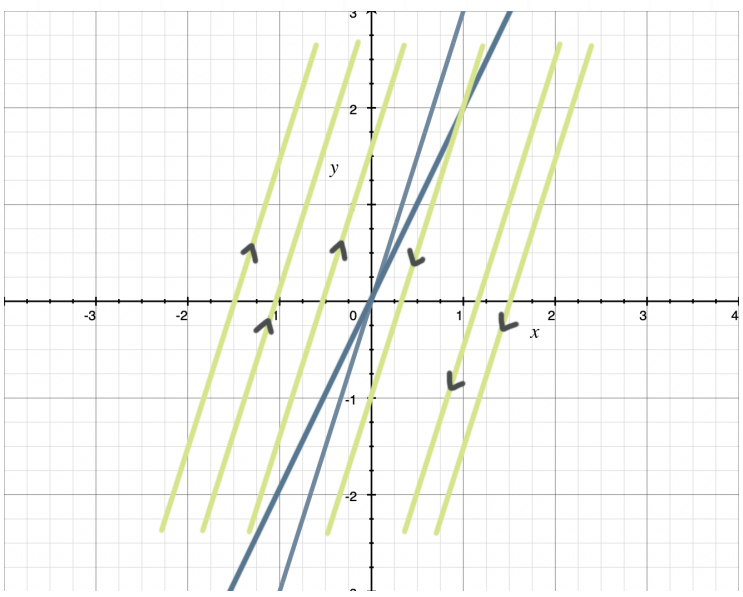
Question: Sketch the phase portrait of the system.

$$x_1' = 3x_1 - x_2$$

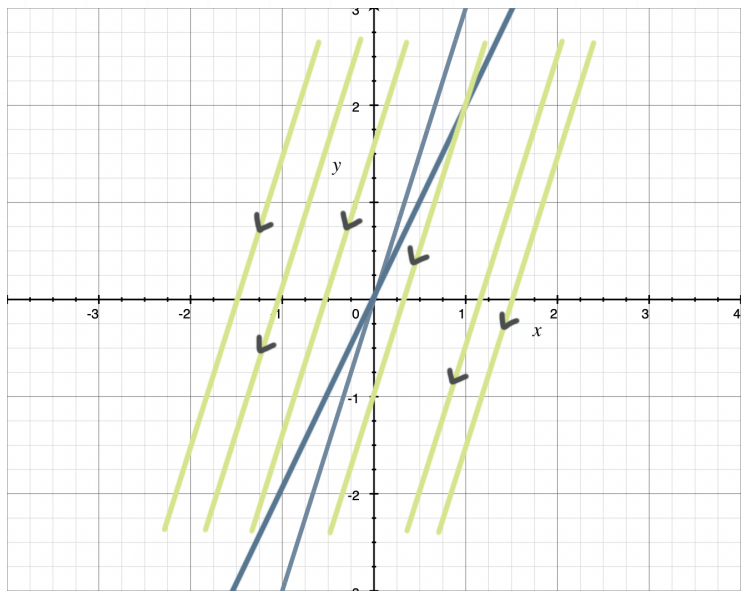
$$x_2' = 9x_1 - 3x_2$$

Answer choices:

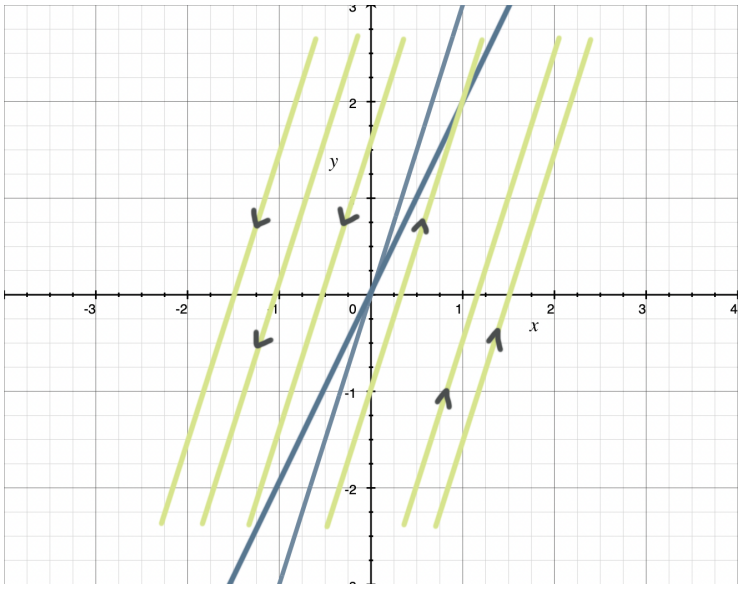
A



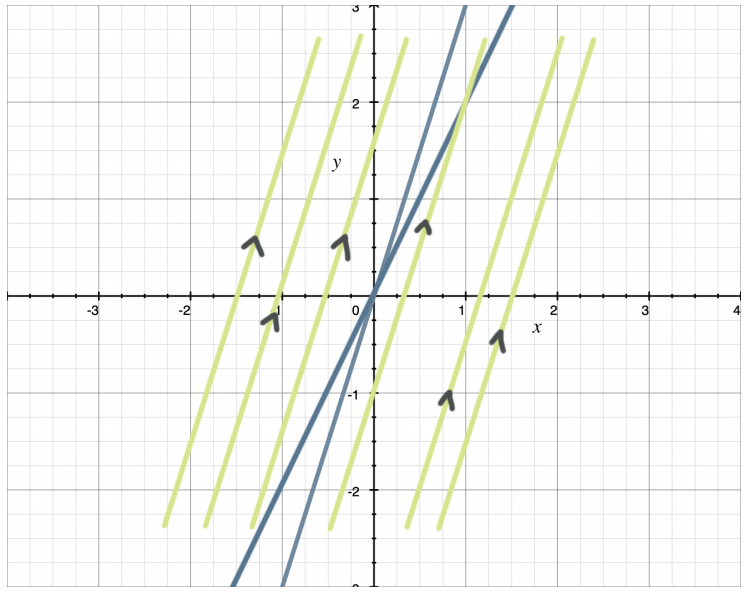
B



C



D



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Solution: C

The coefficient matrix and $A - \lambda I$ are,

$$A = \begin{bmatrix} 3 & -1 \\ 9 & -3 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3 - \lambda & -1 \\ 9 & -3 - \lambda \end{bmatrix}$$

and the characteristic equation

$$(3 - \lambda)(-3 - \lambda) - (-1)(9) = 0$$

$$\lambda^2 = 0$$

$$\lambda = 0$$

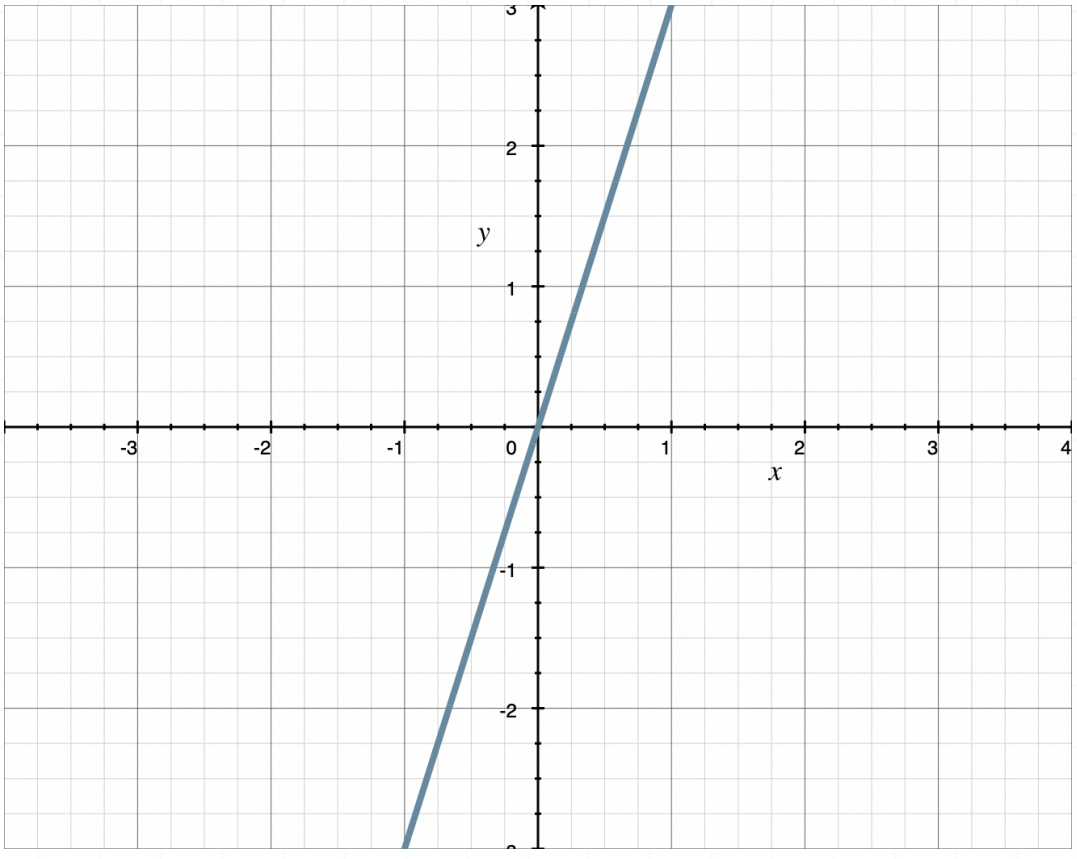
gives the Eigenvalue $\lambda_1 = 0$ and the associated Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1} = (1,3)$ lies along the line $y = 3x$, so we'll sketch this line.

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The Eigenvalue associated with $\overrightarrow{k_1} = (1,3)$ is $\lambda = 0$, and we'll use $\overrightarrow{k_1} = (1,3)$ to find $\overrightarrow{p_1}$.

$$(A - (0)I)\overrightarrow{p_1} = \overrightarrow{k_1}$$

$$\begin{bmatrix} 3 & -1 \\ 9 & -3 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 3p_1 - p_2 \\ 9p_1 - 3p_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Turning this matrix equation into a system of equations gives

$$3p_1 - p_2 = 1$$

$$9p_1 - 3p_2 = 3$$

and then

$$3p_1 - p_2 = 1$$

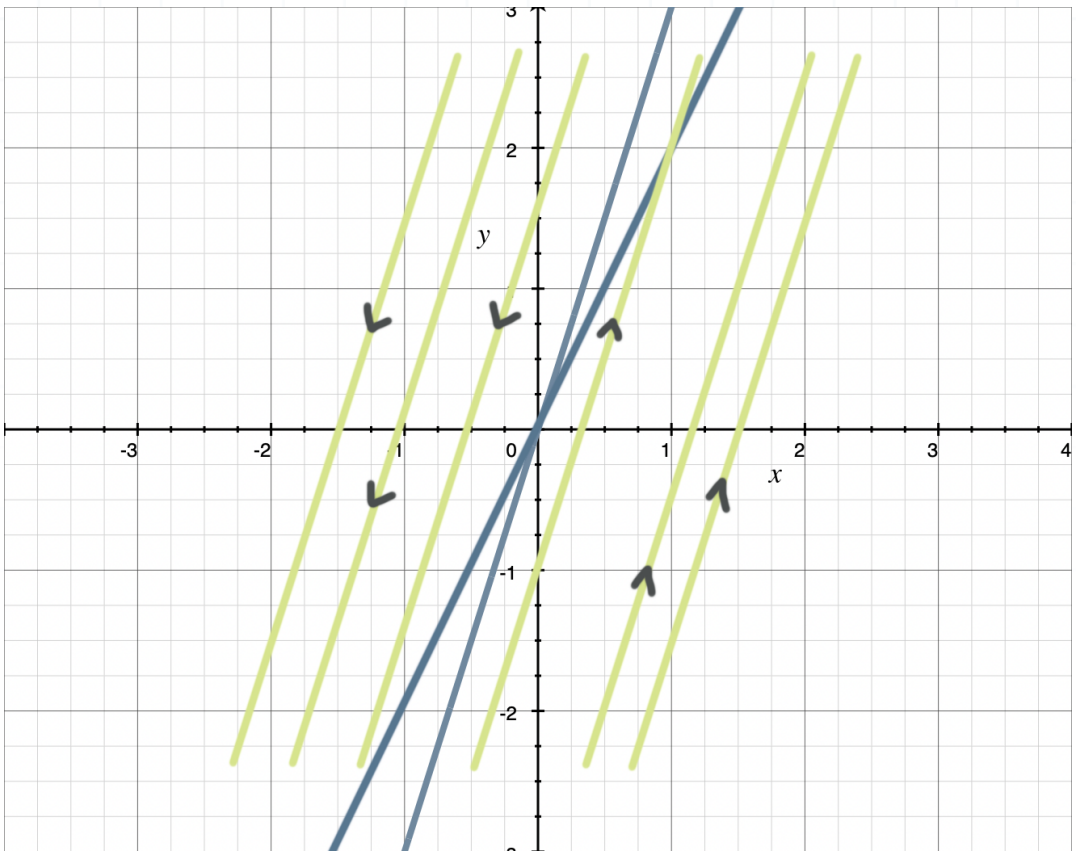
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$$3p_1 = p_2 + 1$$

So we'll use

$$\overrightarrow{p_1} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

The vector $\overrightarrow{p_1} = (1,2)$ lies along the line $y = 2x$. All trajectories are parallel to $\overrightarrow{k_1} = (1,3)$ and every point along $\overrightarrow{p_1} = (1,2)$ is an equilibrium point.



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Topic: Phase portraits for equal real Eigenvalues

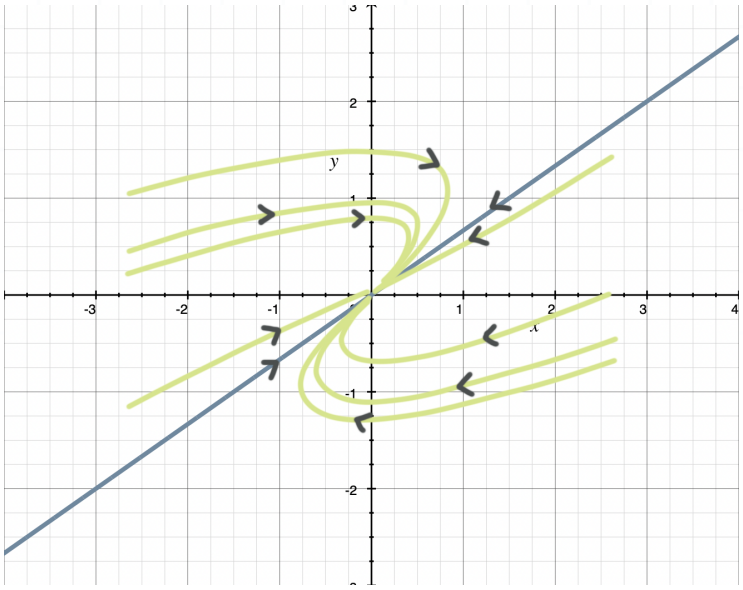
Question: Sketch the phase portrait of the system.

$$x_1' = 12x_1 - 9x_2$$

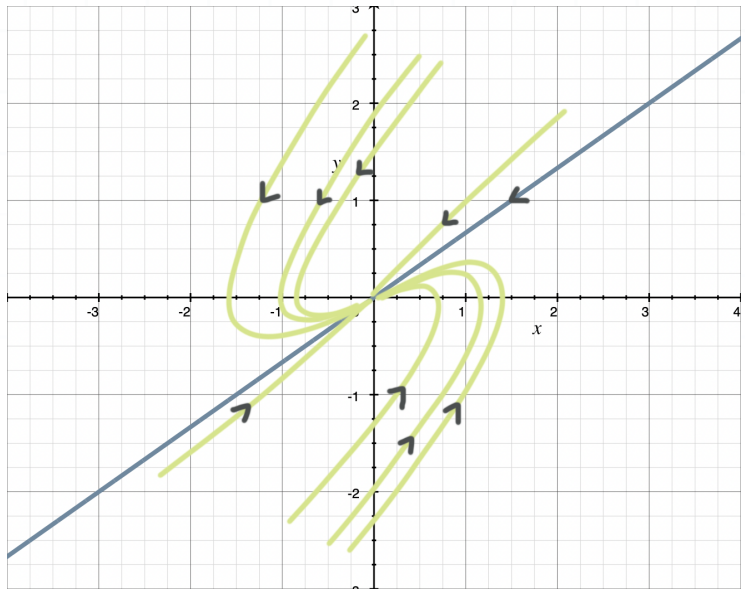
$$x_2' = 4x_1$$

Answer choices:

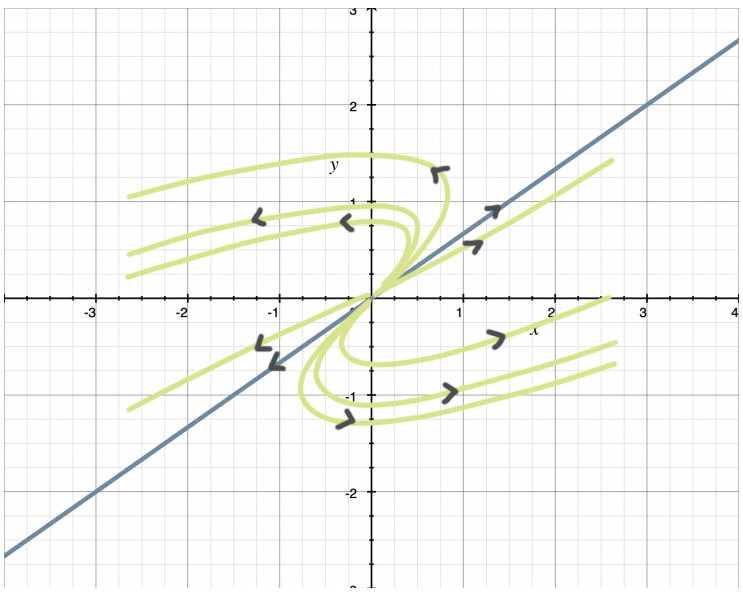
A



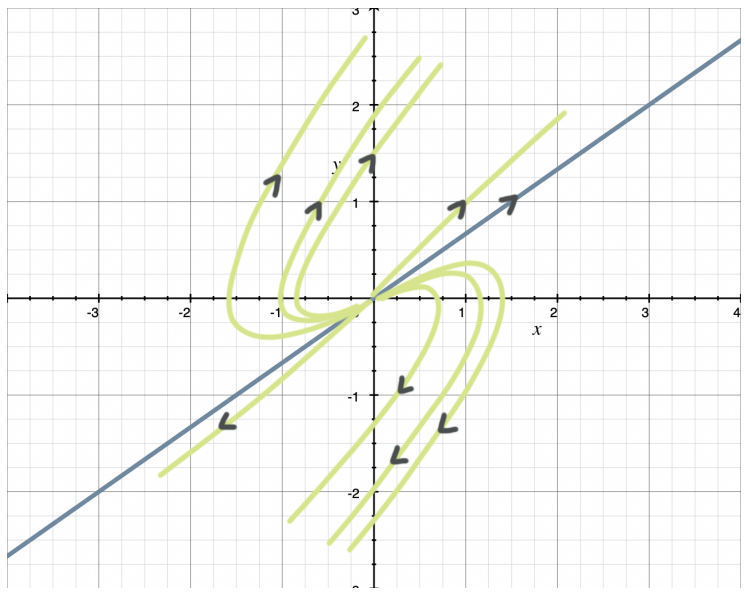
B



C



D



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Solution: C

The coefficient matrix and $A - \lambda I$ are,

$$A = \begin{bmatrix} 12 & -9 \\ 4 & 0 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 12 - \lambda & -9 \\ 4 & -\lambda \end{bmatrix}$$

and the characteristic equation

$$(12 - \lambda)(-\lambda) - (-9)(4) = 0$$

$$\lambda^2 - 12\lambda + 36 = 0$$

$$(\lambda - 6)(\lambda - 6) = 0$$

$$\lambda = 6,\ 6$$

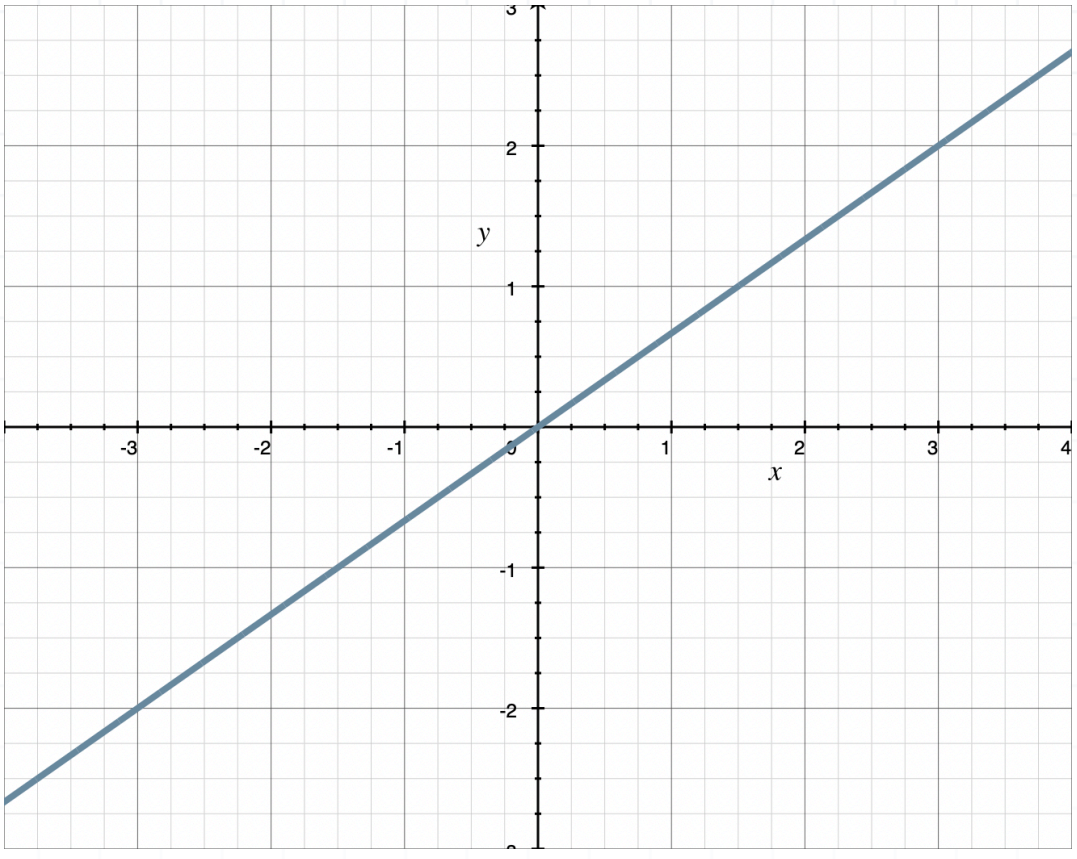
gives the Eigenvalues $\lambda_1 = \lambda_2 = 6$, and the associated Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1} = (3,2)$ lies along the line $y = (2/3)x$, so we'll sketch this line.

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The Eigenvalue associated with $\overrightarrow{k_1} = (3,2)$ is $\lambda = 6$, which means the direction along that trajectory is away from the origin.

Because the Eigenvalue is positive, we're dealing with an unstable repeller node that repels all trajectories.

If we test the vector $\overrightarrow{x} = (1,0)$ in the matrix equation associated with the system, we get

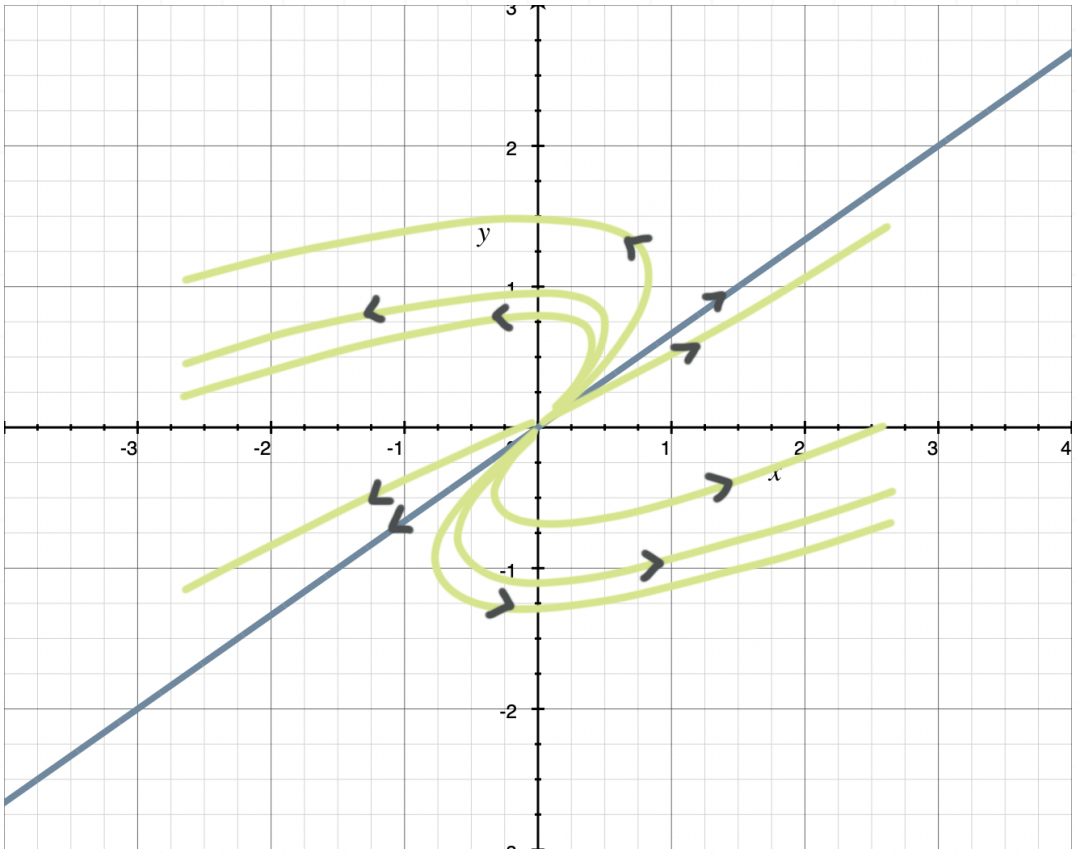
$$\overrightarrow{x}' = \begin{bmatrix} 12 & -9 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\overrightarrow{x}' = \begin{bmatrix} 12(1) - 9(0) \\ 4(1) + 0(0) \end{bmatrix}$$

$$\overrightarrow{x}' = \begin{bmatrix} 12 \\ 4 \end{bmatrix}$$

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This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(12,4)$ (toward the first quadrant), which means that the phase portrait must look something like



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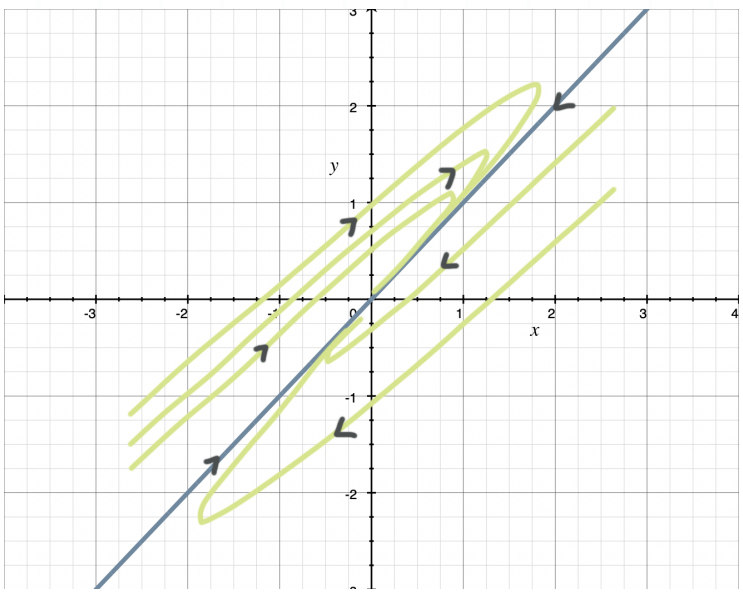
Question: Sketch the phase portrait of the system.

$$x_1' = -6x_1 + 5x_2$$

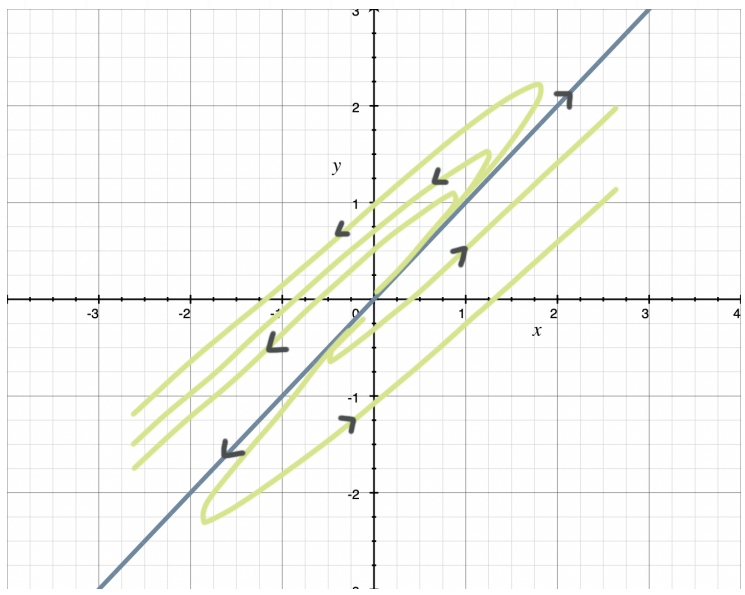
$$x_2' = -5x_1 + 4x_2$$

Answer choices:

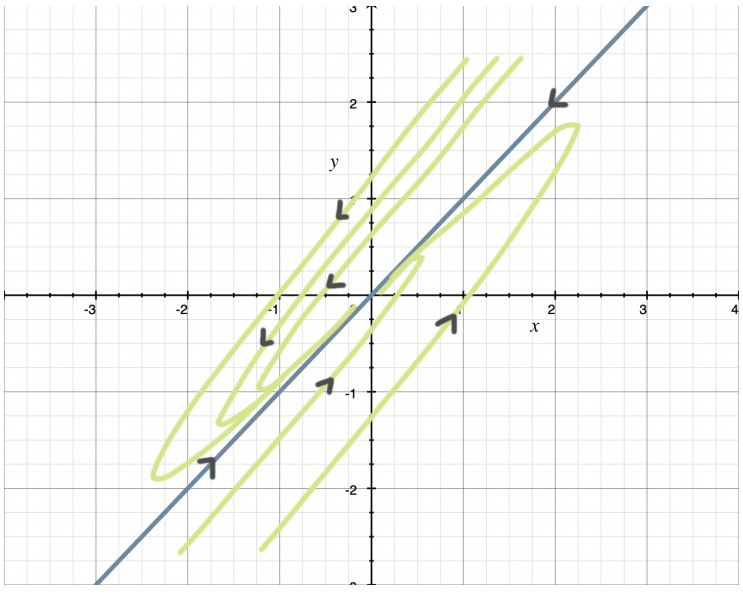
A



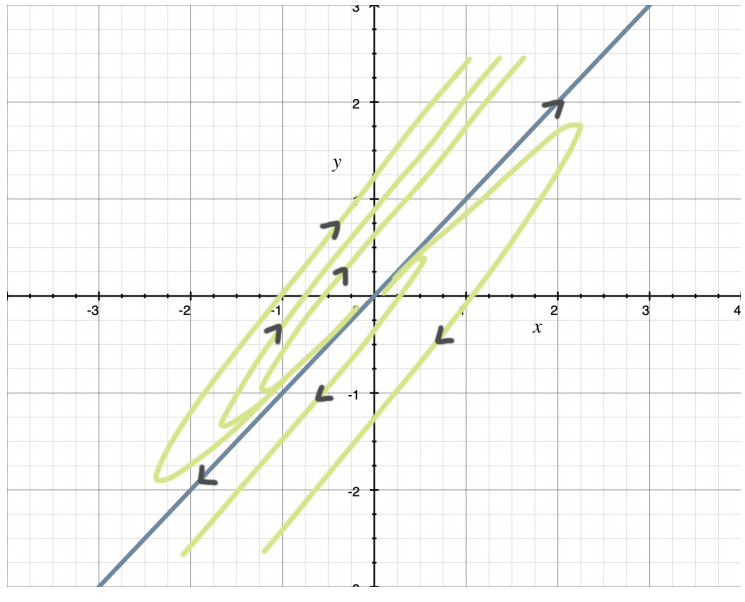
B



C



D



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Solution: A

The coefficient matrix and $A - \lambda I$ are,

$$A = \begin{bmatrix} -6 & 5 \\ -5 & 4 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -6 - \lambda & 5 \\ -5 & 4 - \lambda \end{bmatrix}$$

and the characteristic equation

$$(-6 - \lambda)(4 - \lambda) - (5)(-5) = 0$$

$$\lambda^2 + 2\lambda + 1 = 0$$

$$(\lambda + 1)(\lambda + 1) = 0$$

$$\lambda = -1, \ -1$$

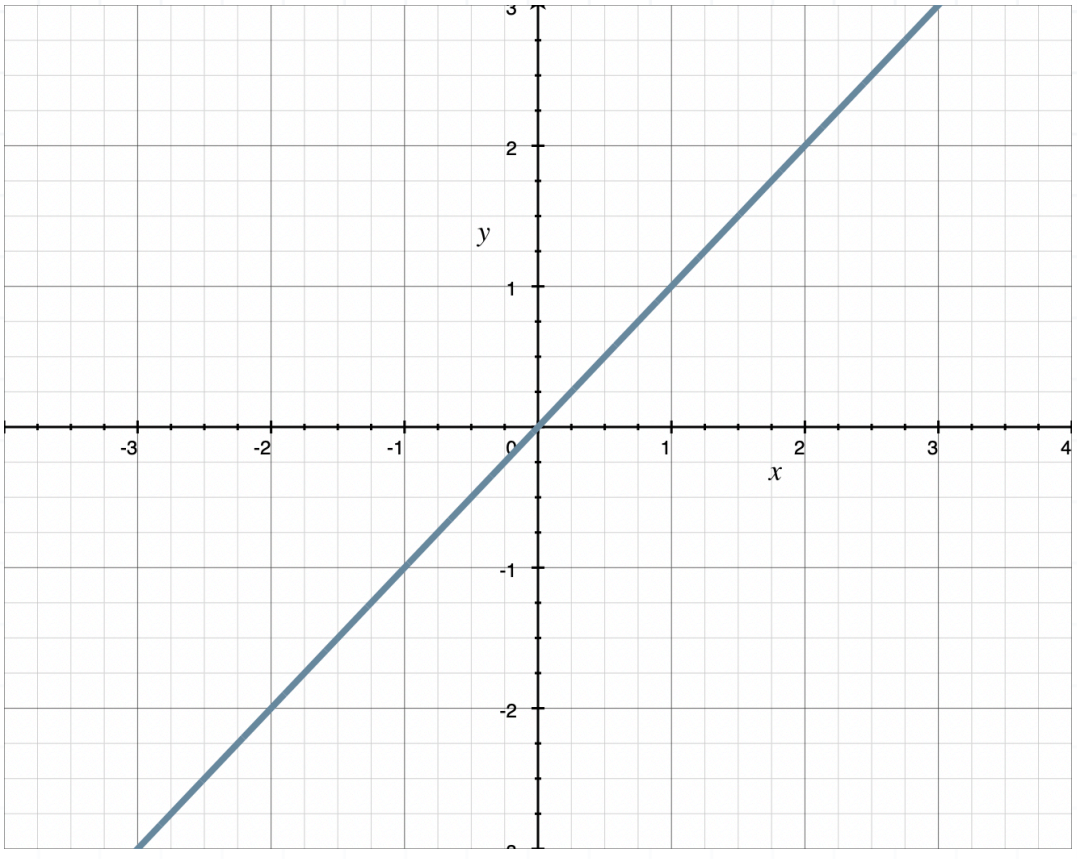
gives the Eigenvalues $\lambda_1 = \lambda_2 = -1$, and the associated Eigenvector

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1} = (1,1)$ lies along the line $y = x$, so we'll sketch this line.

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The Eigenvalue associated with $\overrightarrow{k_1} = (1,1)$ is $\lambda = -1$, which means the direction along that trajectory is toward the origin.

Because the Eigenvalue is negative, we're dealing with an asymptotically stable attractor node that attracts all trajectories.

If we test the vector $\overrightarrow{x} = (1,0)$ in the matrix equation associated with the system, we get

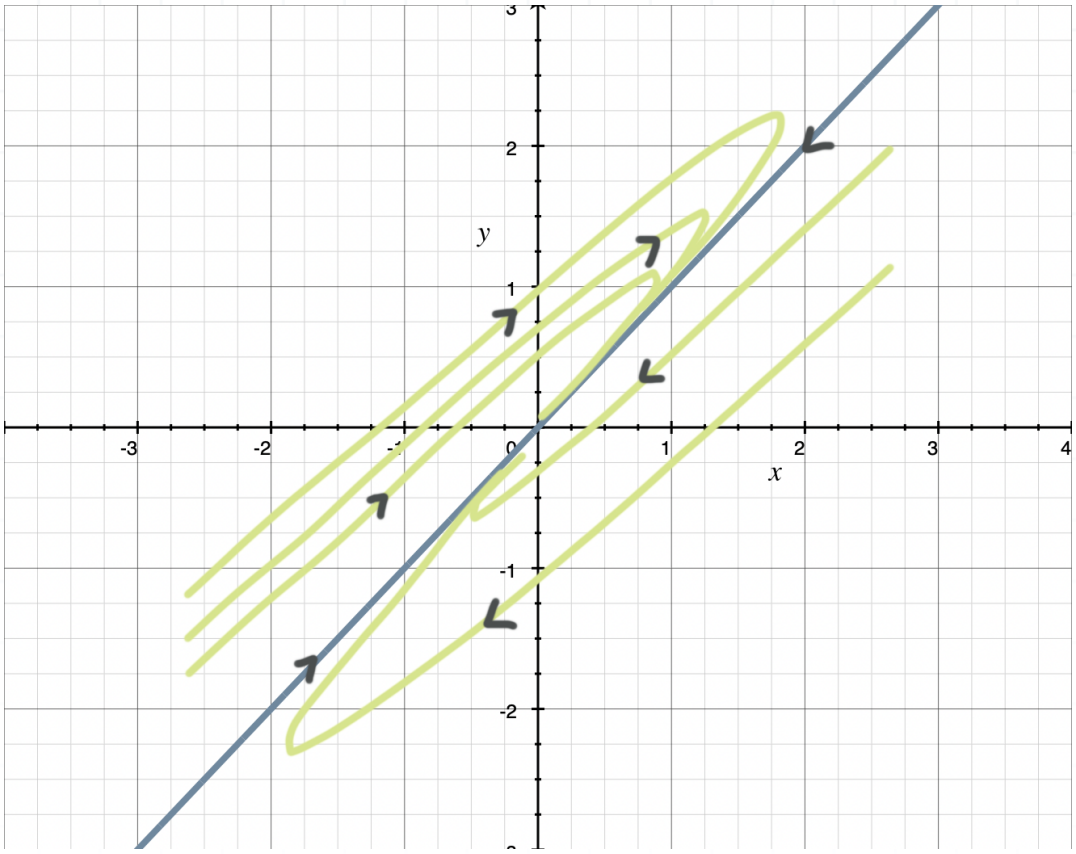
$$\overrightarrow{x}' = \begin{bmatrix} -6 & 5 \\ -5 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\overrightarrow{x}' = \begin{bmatrix} -6(1) + 5(0) \\ -5(1) + 4(0) \end{bmatrix}$$

$$\overrightarrow{x}' = \begin{bmatrix} -6 \\ -5 \end{bmatrix}$$

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This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(-6, -5)$ (toward the third quadrant), which means that the phase portrait must look something like



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